

CPSC 250 - Programming for Data Manipulation

Final Exam

Instructions

You have 150 minutes. This is a closed-book, in-class exam. No access to the internet or external software is allowed. Show all work clearly and legibly.

Part I: Multiple Choice (30 points, 2 points each)

Circle the best answer.

1. What is the output of the following code? `a = [1, 2, 3]`
`b = a`
`b[0] = 99`
`print(a)`
 - A. `[1, 2, 3]`
 - B. `[99, 2, 3]`
 - C. `[1, 99, 3]`
 - D. Error
2. What does `df.dropna()` do in pandas?
 - A. Fills in missing values
 - B. Drops rows or columns with missing values
 - C. Drops duplicate rows
 - D. Drops specified columns
3. What is the purpose of the `__str__` method in a class?
 - A. Initialize the object
 - B. Represent the object as a string

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- D. Compare two objects
4. What is the primary benefit of inheritance in object-oriented programming?
- A. Improved speed
 - B. Data hiding
 - C. Code reuse
 - D. Loop simplification
5. In a recursive function, what is the base case?
- A. The case where an exception is raised
 - B. The function that calls itself
 - C. The condition where recursion stops
 - D. The loop inside the function
6. Which of the following is true about lists and tuples?
- A. Lists are immutable, tuples are mutable
 - B. Both are mutable
 - C. Both are immutable
 - D. Lists are mutable, tuples are immutable
7. What does the `curve_fit` function from `scipy.optimize` do?
- A. Plots data using a curve
 - B. Fits data to a specified function
 - C. Fits a line to data using OLS
 - D. Returns the correlation coefficient
8. What is the time complexity of searching in a balanced binary search tree?
- A. $O(1)$
 - B. $O(n)$
 - C. $O(\log n)$
 - D. $O(n \log n)$
9. Which of the following statements creates a dictionary?
- A. `d = {}`

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- C. `d = ()`
 - D. `d = set()`
10. What happens when you override the `__eq__` method in a class?
 - A. You redefine how two instances are compared with `==`
 - B. You make your class hashable
 - C. You initialize the object
 - D. You compare types of objects
 11. What does the pandas method `read_csv()` return?
 - A. A list of strings
 - B. A NumPy array
 - C. A Series
 - D. A DataFrame
 12. What does operator overloading allow in Python?
 - A. Redefining loops
 - B. Creating multiple classes in one file
 - C. Using operators like `+` with user-defined types
 - D. Automatically generating documentation
 13. In pandas, what does `df['column'].mean()` compute?
 - A. The median
 - B. The total
 - C. The standard deviation
 - D. The average
 14. What is polymorphism in OOP?
 - A. The ability of different classes to share the same attributes
 - B. The ability of objects to take on multiple forms via shared methods
 - C. Hiding data inside an object
 - D. Changing variable types at runtime
 15. What is the correct way to inherit from a class in Python?

- A. class A inherits B
- B. class A: inherits B
- C. class A(B):
- D. class A:

Part II: Error Identification (15 points) CPSC 250 - Final Exam

The following code is meant to reverse a list recursively. Identify and correct all errors.

```
def reverse_list(lst):  
    if len(lst) = 0:  
        return []  
    else:  
        return reverse_list(lst[1:]) + lst[0]
```

Part III: Code Writing (30 points)

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Q1. (6 points) Write a function `is_palindrome(s)` that returns `True` if the string `s` is a palindrome, and `False` otherwise.

Q2. (8 points) Write a class `Student` with fields for `name` and `gpa`, and a `__str__` method that prints the student's name and GPA. Also include an `__lt__` method that compares students by GPA.

Q3. (8 points) Write a function `load_and_filter(filename)` that reads a CSV file into a pandas `DataFrame`, and returns only the rows where the value in the column "score" is greater than 80.

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Q4. (8 points) Write a base class `Animal` with a method `make_sound()` that returns the string "...". Create two derived classes `Dog` and `Cat` that override `make_sound()` to return "woof" and "meow", respectively. Show how polymorphism allows you to loop over a list of animals and call `make_sound()` on each.

Part IV: Code Comprehension and Commenting (25 points)

This function is designed to perform curve fitting using `curve_fit`. Comment each line and explain what the code is doing overall.

```
import numpy as np
from scipy.optimize import curve_fit

def model(x, a, b):
    return a * np.exp(b * x)

def fit_data(x, y):
    popt, pcov = curve_fit(model, x, y)
    print("a=", popt[0])
    print("b=", popt[1])
```

What do `popt` and `pcov` represent in this context?